

Medical Insurance Cost Prediction Using Machine Learning

Project Report

Submitted By

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Internship: Data Science Internship

Project Title: Medical Insurance Cost Prediction Using Machine Learning

Tools Used: Python, Scikit-learn, Streamlit, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

Medical insurance companies estimate insurance charges based on various personal and health-related factors such as age, Body Mass Index (BMI), smoking habits, number of children, and residential region.

Accurately predicting insurance charges helps insurance providers make better pricing decisions and allows customers to estimate their expected medical expenses. This project develops a machine learning model that predicts medical insurance charges based on user inputs and deploys the model through an interactive Streamlit web application.

2. Problem Statement

Insurance companies use multiple customer attributes to determine insurance premiums. Manually estimating these costs can be difficult due to the complex relationship between various features.

The objective of this project is to build a machine learning model capable of predicting insurance charges accurately and deploy the model as a user-friendly web application.

3. Objectives

The objectives of this project are:

- Perform data preprocessing and cleaning.
 - Analyze relationships among different features.
 - Train multiple regression models.
 - Compare model performance using evaluation metrics.
 - Improve the best-performing model through hyperparameter tuning.
 - Deploy the trained model using Streamlit.
 - Provide an interactive interface for real-time predictions.
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4. Dataset Description

The Medical Insurance dataset contains information about individuals and their corresponding insurance charges.

Dataset Information

Attribute	Description
Records	2772
Features	6
Target Variable	Insurance Charges

Features Used

- Age
- Gender
- BMI
- Number of Children
- Smoking Status
- Region

Target Variable

- Insurance Charges
-

5. Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Streamlit
 - Matplotlib
 - Seaborn
 - Joblib
 - Git & GitHub
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6. Data Preprocessing

The following preprocessing steps were performed:

- Checked missing values.
- Verified duplicate records.
- Encoded categorical variables into numerical format.
- Split the dataset into training and testing sets.
- Prepared the data for machine learning models.

No missing values were present in the dataset.

7. Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset.

The analysis included:

- Distribution of BMI
- Correlation Heatmap
- Feature Importance
- Model Comparison
- Baseline vs Tuned Model Performance

These visualizations helped identify important relationships between variables.

8. Machine Learning Models

The following regression algorithms were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

Each model was trained using the same training dataset and evaluated using identical testing data.

9. Model Evaluation

The models were evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Random Forest Regressor achieved the best overall performance and was selected as the final model.

10. Hyperparameter Tuning

Grid Search Cross Validation was used to optimize the Random Forest model.

The following parameters were tuned:

- Number of Estimators
- Maximum Depth
- Minimum Samples Split
- Minimum Samples Leaf

Best Parameters

```
n_estimators = 200  
max_depth = 20  
min_samples_split = 2
```

```
min_samples_leaf = 1
```

The tuned model improved prediction accuracy while reducing prediction error.

11. Model Deployment

The trained Random Forest model was saved using Joblib/Pickle and deployed using Streamlit.

The application allows users to:

- Select Age
- Choose Gender
- Enter BMI
- Select Number of Children
- Choose Smoking Status
- Select Region

After clicking the **Predict Insurance Cost** button, the model estimates the expected insurance charges and displays a prediction summary.

The application was successfully deployed on Streamlit Community Cloud.

12. Results

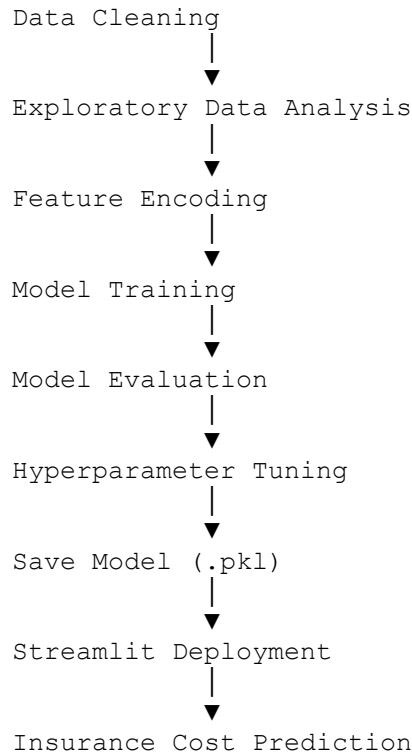
The deployed application successfully predicts insurance charges based on user inputs.

The web application provides:

- Interactive user interface
 - Real-time predictions
 - Prediction summary
 - Clean and responsive layout
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13. Project Workflow

Medical Insurance Dataset
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14. Challenges Faced

During the project, several challenges were encountered:

- Selecting the best-performing regression model.
- Optimizing hyperparameters using Grid Search.
- Saving and loading the trained model.
- Deploying the application using Streamlit Community Cloud.
- Managing project dependencies for deployment.

These challenges were successfully resolved through experimentation and testing.

15. Future Improvements

Possible improvements include:

- Using larger insurance datasets.
- Implementing XGBoost and LightGBM models.
- Adding feature importance explanations for each prediction.
- Deploying the application using Docker and cloud platforms.

- Integrating the application with an online database.

16. Conclusion

This project successfully developed a machine learning system capable of predicting medical insurance charges using customer information.

Multiple regression models were evaluated, and the Random Forest Regressor was selected as the best-performing model after hyperparameter tuning.

The trained model was successfully deployed as an interactive Streamlit web application, allowing users to obtain real-time insurance cost predictions through a simple web interface.

The project demonstrates the complete end-to-end machine learning pipeline, including data preprocessing, exploratory analysis, model development, evaluation, optimization, and deployment.

References

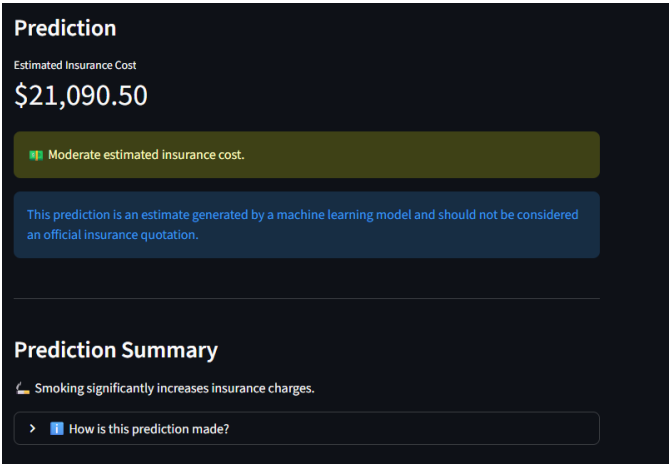
1. Scikit-learn Documentation
2. Streamlit Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Medical Insurance Dataset (Kaggle)

Appendices

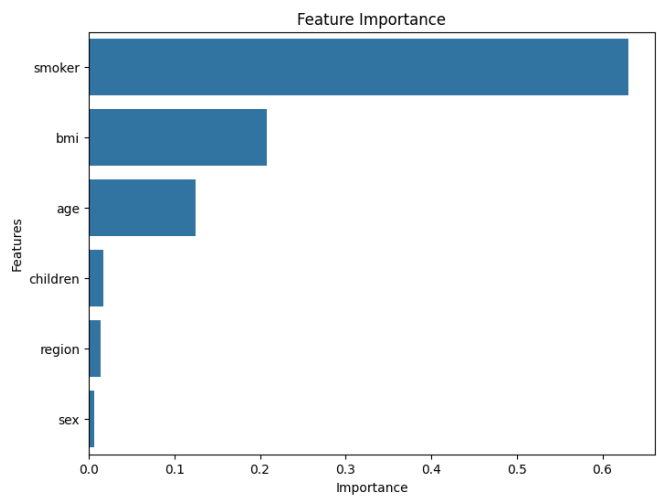
1. Home Page

The screenshot shows the home page of a Streamlit web application titled "Medical Insurance Cost Prediction". On the left is a dark sidebar with a menu containing: "About", "Model" (with sub-item "Random Forest Regressor"), "Dataset" (with sub-item "Medical Insurance Dataset"), "Target" (with sub-item "Insurance Charges"), "Author" (with sub-item "Zaara Khan" and "B.Tech CSE Student"). The main content area has a light blue header with a medical icon and the title. Below the title is a description: "Predict the estimated medical insurance charges based on a person's details using a Random Forest Machine Learning Model." and a prompt: "Fill in the information below and click Predict Insurance Cost." There are three input fields: "Select Age" with a slider set to 30, "Select Gender" with a dropdown menu showing "Female", and "Select BMI" with a slider set to 25.00. At the bottom of the input area is a blue button labeled "Predict Insurance Cost".

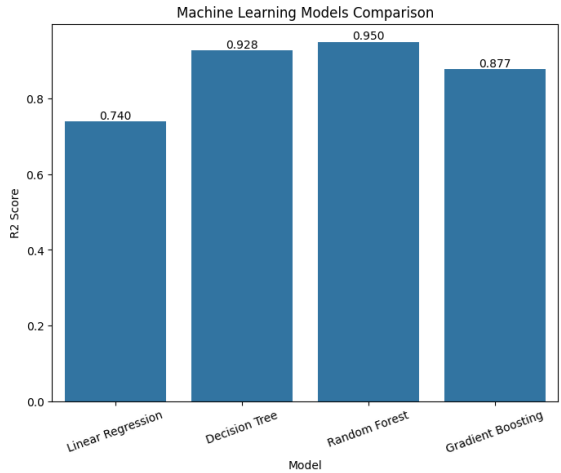
2. Prediction



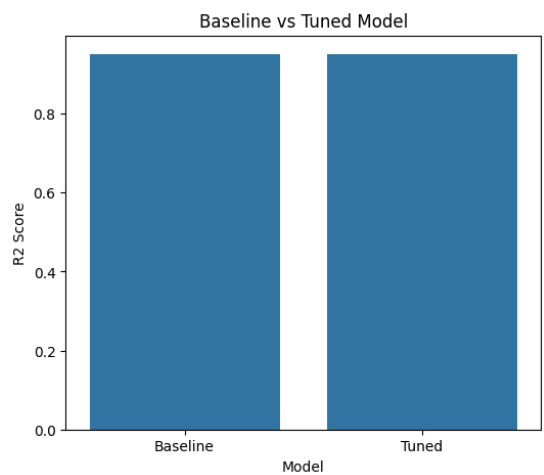
3. Feature Importance



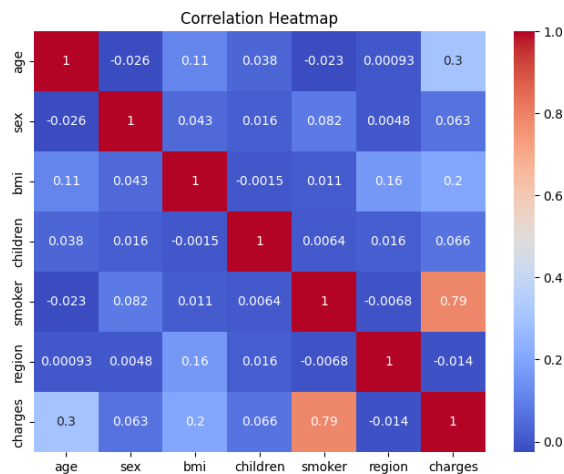
4. Model Comparison



5. Tuned vs Baseline



6. Correlation Heatmap



7. BMI Distribution

